

Muhammad Huzaifa

AI / LLM Engineer

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AI/LLM engineer with 3+ years on systems that are actually in production rather than demos. Day to day that means routing requests across Anthropic and OpenAI model tiers, agentic pipelines with a critic step and a retry loop, RAG and semantic search over 50,000+ documents a month, and keeping token cost under control. I build the full stack around those systems too, Python (Django, FastAPI) and TypeScript (Next.js, NestJS). Two published papers, and 3rd Prize in the AI Track at the Solana Breakout Hackathon (\$15,000, from 8,300+ entries).

PROFESSIONAL EXPERIENCE

Rehani Soko LLC • Remote (US company)

Apr 2023 – Present

Software Developer (Full-Time since Nov 2024; Contract Apr 2023 – Oct 2024)

- Build and run the LLM side of the platform. Requests route across Anthropic and OpenAI model tiers depending on what the task actually needs, and I count tokens with tiktoken so I know what each feature costs to run.
- Built a content pipeline that checks its own work. A critic model scores every generated draft, rejects the weak ones, and sends them back to be regenerated before anything publishes. That took manual review out of the loop entirely.
- Shipped the WhatsApp and web assistants. They keep a long-term profile of each user's preferences, pulled out of past conversations and merged over time, and handle voice through Whisper and TTS.
- Built semantic matching on text-embedding-3-small, and an underwriting service where the model has to return structured output that passes schema validation before anything downstream will touch it.
- Wrote the scrapers and cleaning pipeline behind all of it, roughly 50,000 listings a month. Reliability improved about 35% after the consistency checks went in.
- Also lead feature work on the Vue.js and Django platform, and built the React/Vite admin panel the content team uses.

Agent Arc • Remote (Singapore)

Jan 2025 – Jul 2025

Frontend Lead Developer

- Led frontend for a non-custodial AI-driven trading terminal on Solana; integrated AI/ML trading APIs into the execution flow.
- Implemented Privy authentication and wallet integration, and on-chain fee handling via staking contract ABIs.

Lovum Group • Remote (US company)

Jan 2024 – Dec 2025

Software Engineer (Part-Time)

- Worked on two of their products, EduEssential and Waridi, building the Next.js frontends and the Python APIs behind them, and getting both to scale.
- Designed and wired up the APIs, including the endpoints feeding their ML features.

Algorithm Consulting (Pvt) Ltd • Karachi, Pakistan

May 2024 – Oct 2024

Junior Developer

- Worked on an ASP.NET enterprise application, mostly new features and the bug queue. A lot of it was pairing with senior engineers on query performance and release stability.

CrossWing Inc. • Remote (Ontario, Canada)

Mar 2023 – Nov 2023

Full-Stack Developer

- Built and maintained the React.js web apps that sat on top of their robotics products.
- Collaborated with the AI/ML engineers and product managers on features connecting the hardware to the web side.

Profecia Links Consulting Pvt. Ltd. • Remote (Abu Dhabi, UAE)

May 2022 – Aug 2022

Software Engineer (A1)

- Automated client data-extraction workflows using Import.io, which cut out a lot of manual collection.

AWARDS & PUBLICATIONS

- **3rd Prize, AI Track (\$15,000), Solana Breakout Hackathon (Global).** Selected among 8,300+ submissions for Agent Arc, a non-custodial AI-driven trading terminal on Solana.
- Huzaifa, M. et al. (2023). *An Integrated Approach to Robotic Object Grasping and Manipulation using 6D Pose Estimation.* arXiv:2411.13205.
- Huzaifa, M. et al. (2025). *Smart Shelf Advertising using Real-Time Product Segmentation and Interaction on Low-Cost Edge Devices.* DOI:10.13140/RG.2.2.34475.96800.

SELECTED PROJECTS

F1GPT · Formula One RAG Assistant

Aug 2025

Next.js 15, RAG, LangChain, Astra DB, Puppeteer, Vercel AI SDK

- A Retrieval-Augmented Generation assistant for F1 race stats and rules.
- Wrote the scrapers and chunking pipeline, stored embeddings in Astra DB, and added a streaming LLM layer with tool use so answers come back cited.

LLM Semantic Book Recommender

Jun 2025

LangChain, Vector DB, Embeddings, LLM APIs

- Indexes books into a vector database and retrieves them semantically, combining dense search with metadata filters. An LLM re-ranks the results and explains why it picked each one.

SmartScan Billing Assistant

May 2024 – Jul 2024

YOLOv8, Computer Vision, Object Detection & Tracking, Python

- A checkout system that recognises products from a camera feed and adds up the bill. Ended up deployed across 1,200+ products in a working supermarket.

Also shipped: cross-platform mobile (React Native / Expo)

Jan 2026 – Aug 2026

- **Wash42** (car-wash booking & subscriptions) and **Dugan Psychiatry** (mental-health platform with Socket.IO chat, WebRTC video consultations, and Stripe payments) · both released to the App Store and Play Store via an EAS Build/Submit pipeline. Live at apps.apple.com/us/app/wash42/id6803226300, play.google.com/store/apps/details?id=com.wash42.app and play.google.com/store/apps/details?id=com.duganpsychiatry.app

TECHNICAL SKILLS

AI/LLM: RAG, LangChain, Agents & Tool Use, Prompt Engineering, Structured Output, Embeddings & Vector Search (Astra DB), Multi-Provider Routing (Anthropic Claude, OpenAI), Token & Cost Accounting (tiktoken), Streaming, Whisper STT / TTS

Languages: Python, JavaScript/TypeScript (ES6+), C#

Backend: FastAPI, Django, NestJS, Node.js, Express.js, Flask, ASP.NET, REST, GraphQL, Celery

Frontend: React.js, Next.js, Vue.js, Nuxt, Tailwind CSS

ML & Data: Scikit-learn, Pandas, NumPy, YOLOv8, OpenCV, ETL, Data Pipelines, Selenium, Puppeteer, BeautifulSoup

Databases: PostgreSQL, MySQL, MongoDB, Firestore

Mobile & DevOps: React Native, Expo, EAS Build/Submit, WebRTC, Docker, GitHub Actions, Git

EDUCATION

NED University of Engineering & Technology, B.E.

2019 – 2023 | Karachi, Pakistan

Computer & Information Systems Engineering. GPA 3.289/4.0. Coursework: Data Structures & Algorithms, Software Engineering, Artificial Intelligence, Machine Learning.